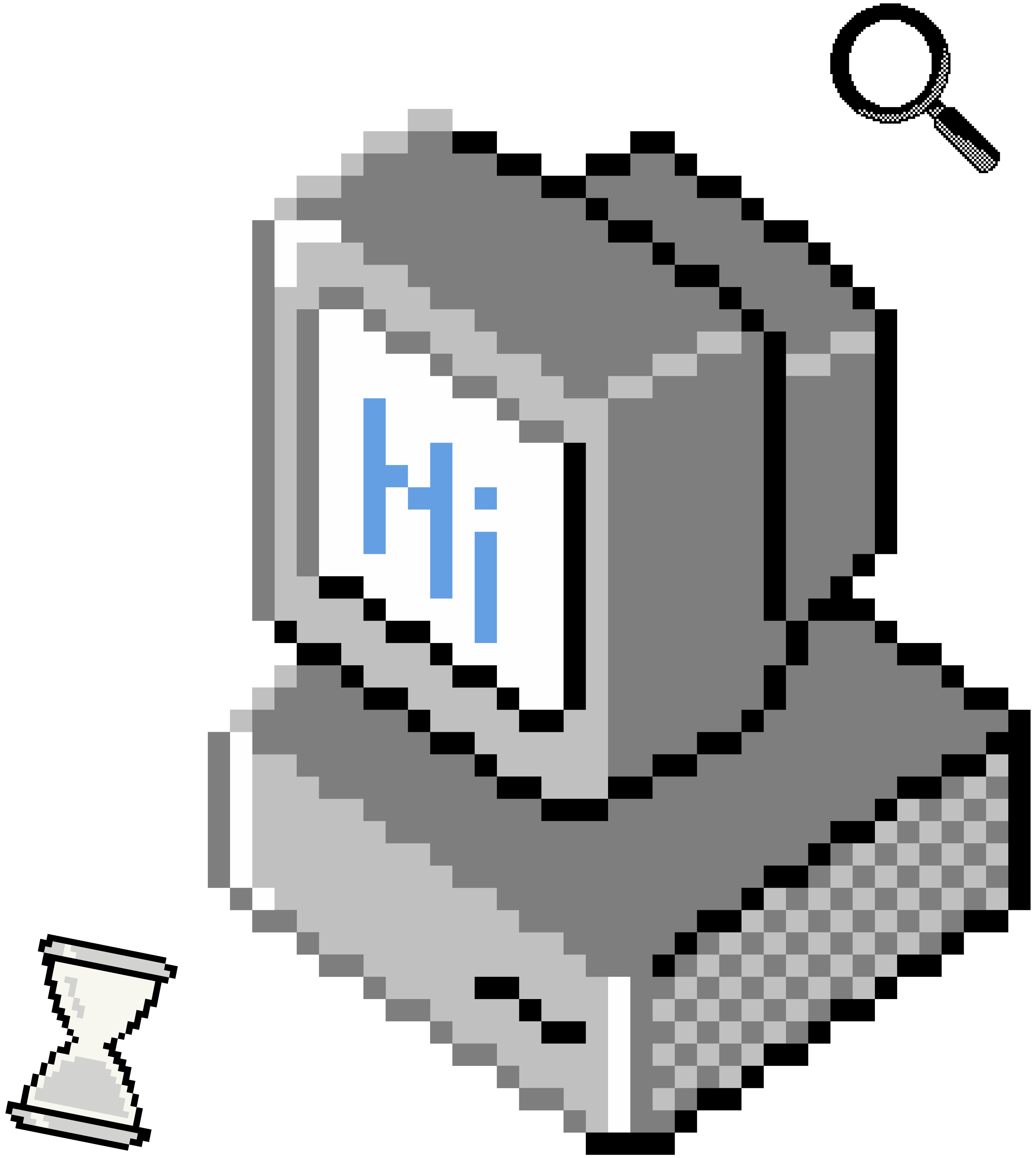


Final Year Project

REAL-TIME HAND GESTURE RECOGNITION FOR GAME CONTROL

Ainul Mardhiah binti Azwan

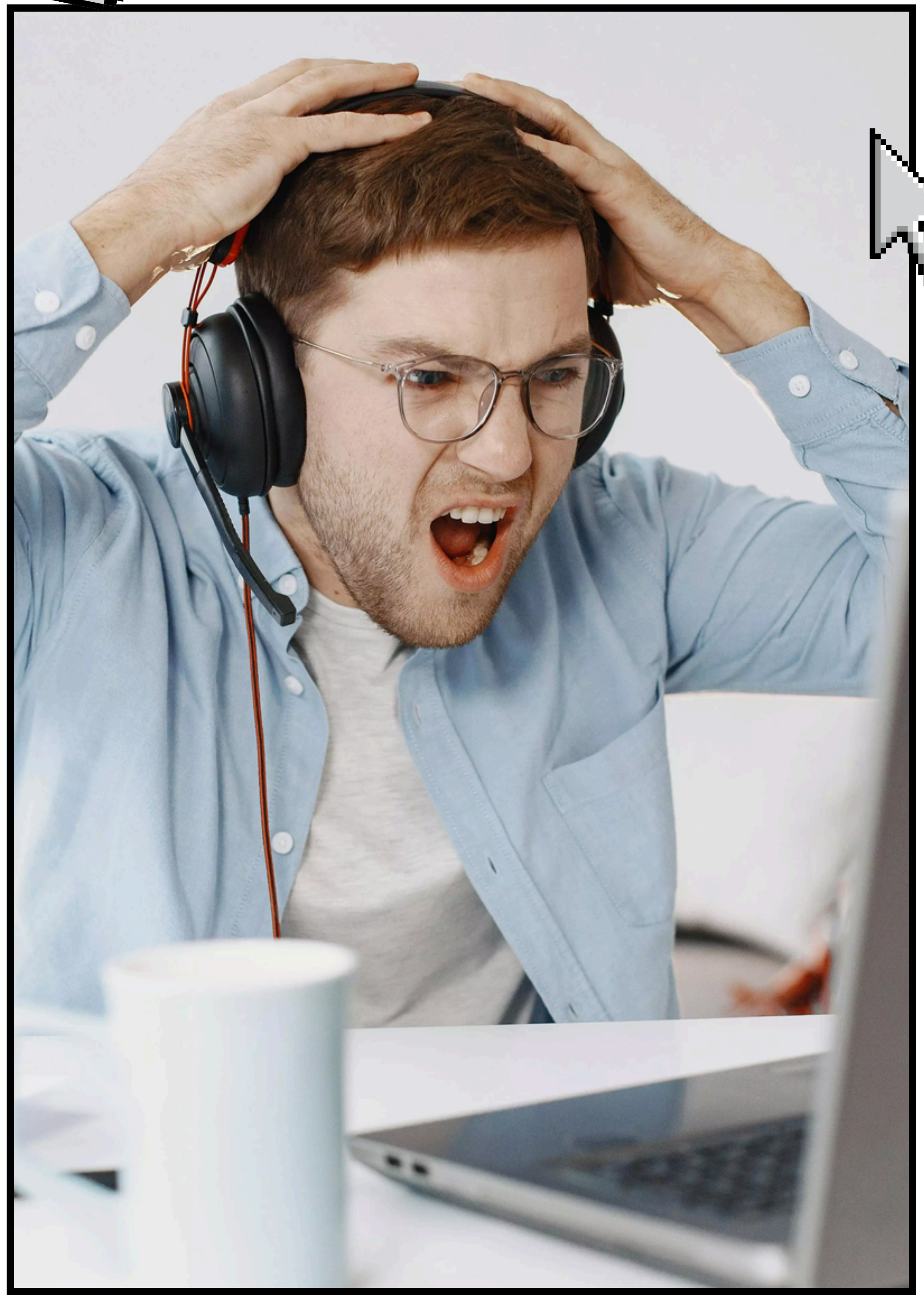
PROBLEM STATEMENT

Pain point:

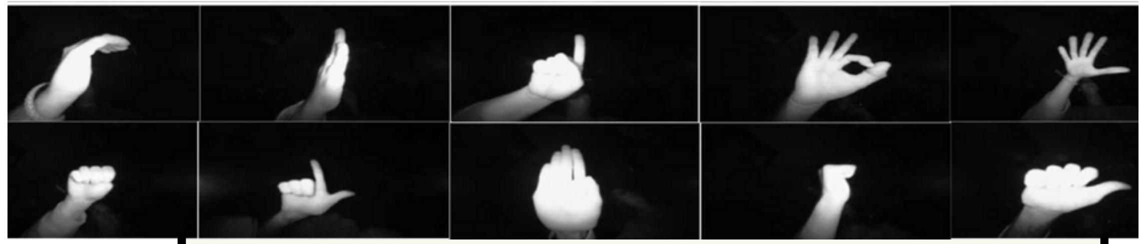
- Traditional game controls requires keyboard/controllers
- Not accessible for everyone

Stakeholders:

- Gamers
- Accessibility users



DATA OVERVIEW



LeapGestRecog

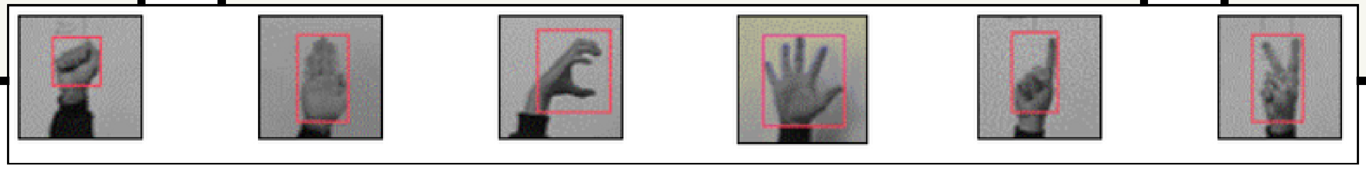
- source: Kaggle
- classes: thumb, fist, palm, index
- size: ~20,000 imgs

Marcel-Train

- source: Idiap.ch
- classes: fist, palm, index, V
- size: ~5,000 imgs

Self-Collected dataset

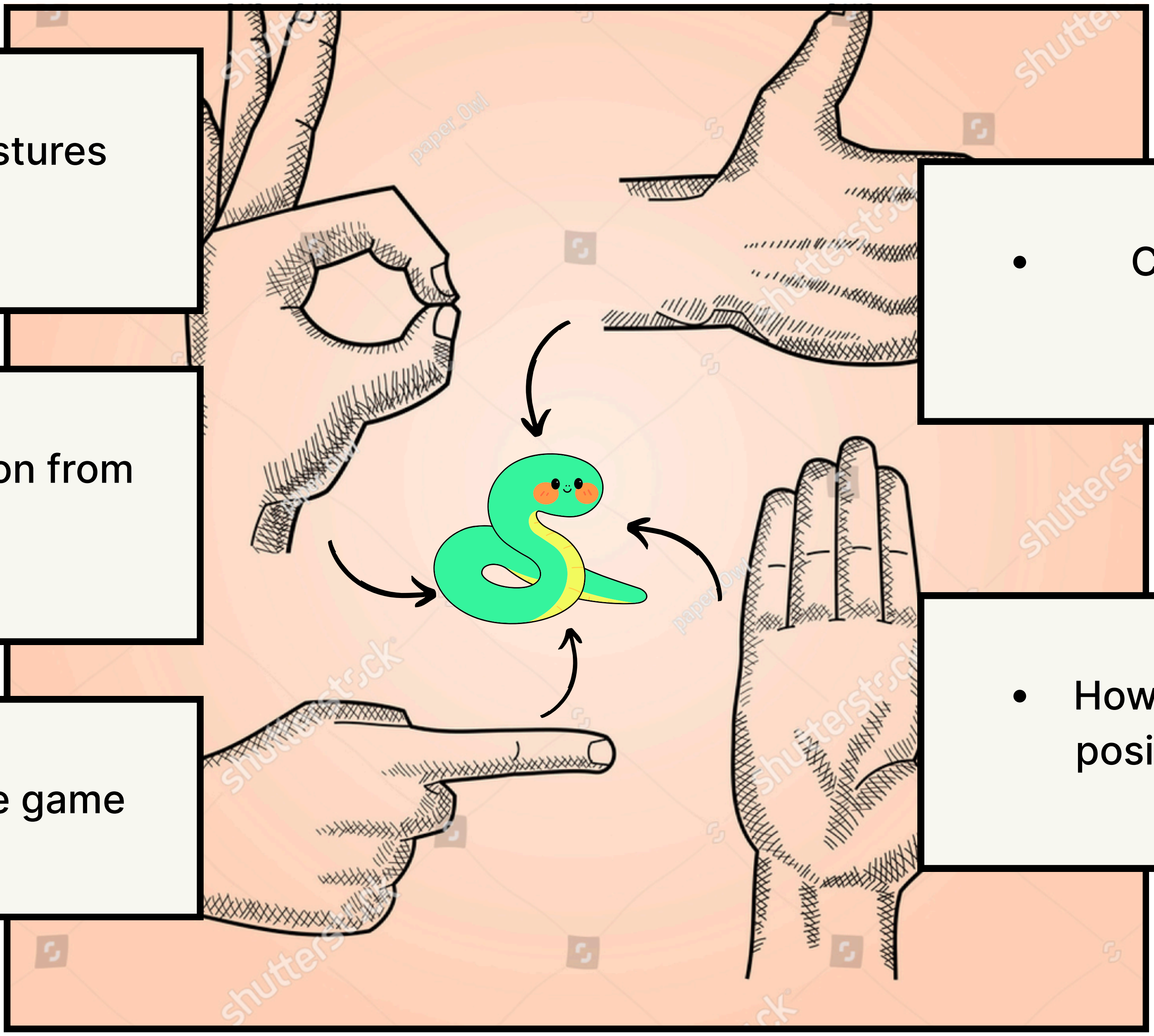
- source: friends, myself
- classes: fist, palm, left, right
- size: ~2,500 imgs



Final Dataset

OBJECTIVES

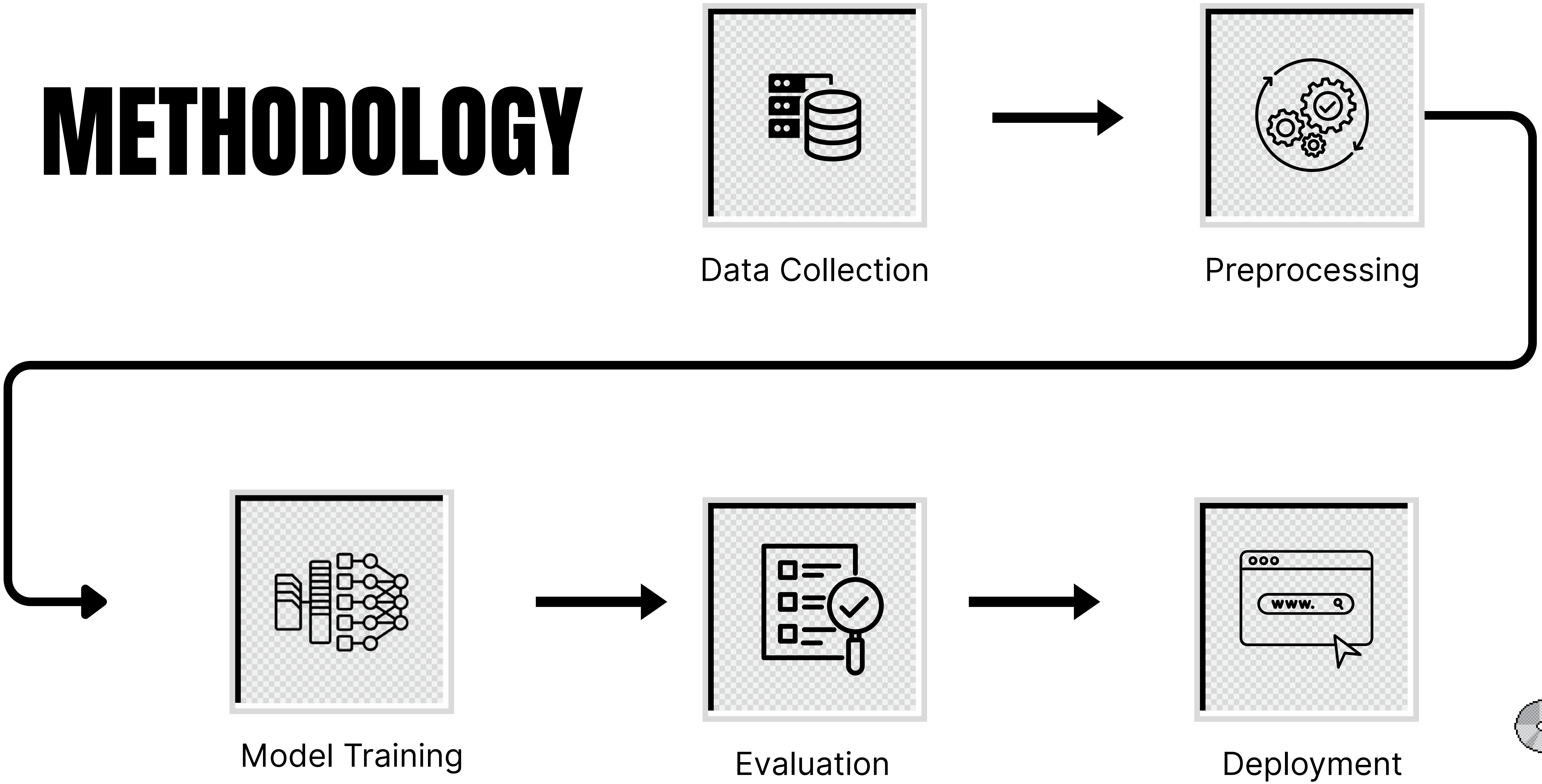
- Train a CNN to classify hand gestures accurately
- Enable real-time gesture prediction from webcam
- Integrate predictions into a snake game



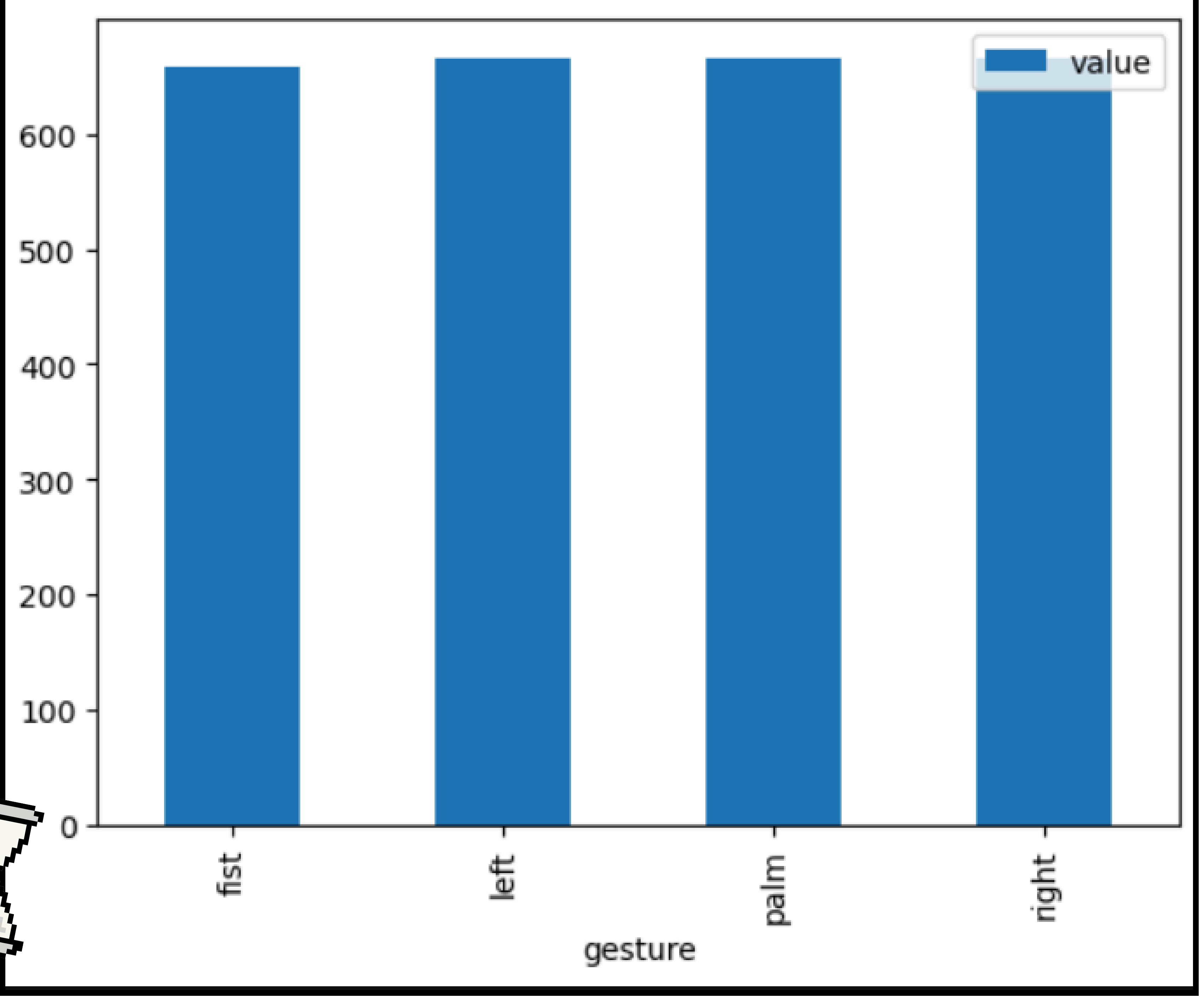
KEY QUESTIONS

- Can a CNN generalize to unseen environments?
- How does background, light, and hand position variation affect performance?

METHODOLOGY



```
Gesture: fist, Number of Images: 659
Gesture: left, Number of Images: 665
Gesture: palm, Number of Images: 665
Gesture: right, Number of Images: 666
<Axes: xlabel='gesture'>
```



EDA KEY FINDINGS (1/3)

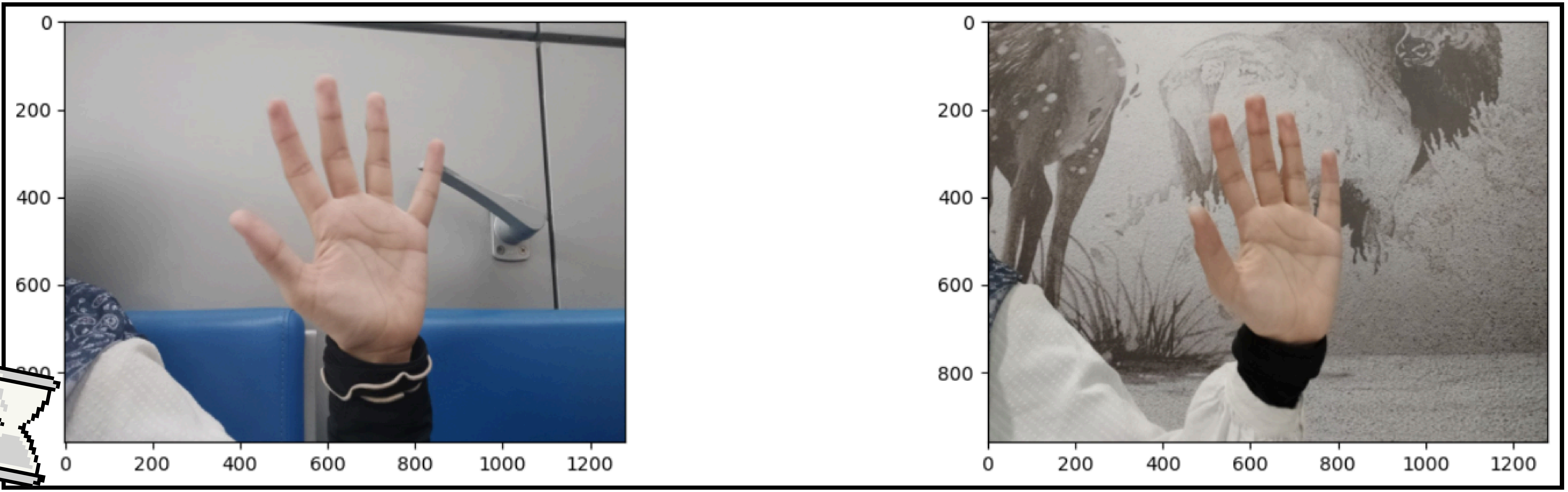
Bar chart of image counts per gesture

- classes are generally evenly balanced
- 2655 total images

EDA KEY FINDINGS (2/3)

Background variation

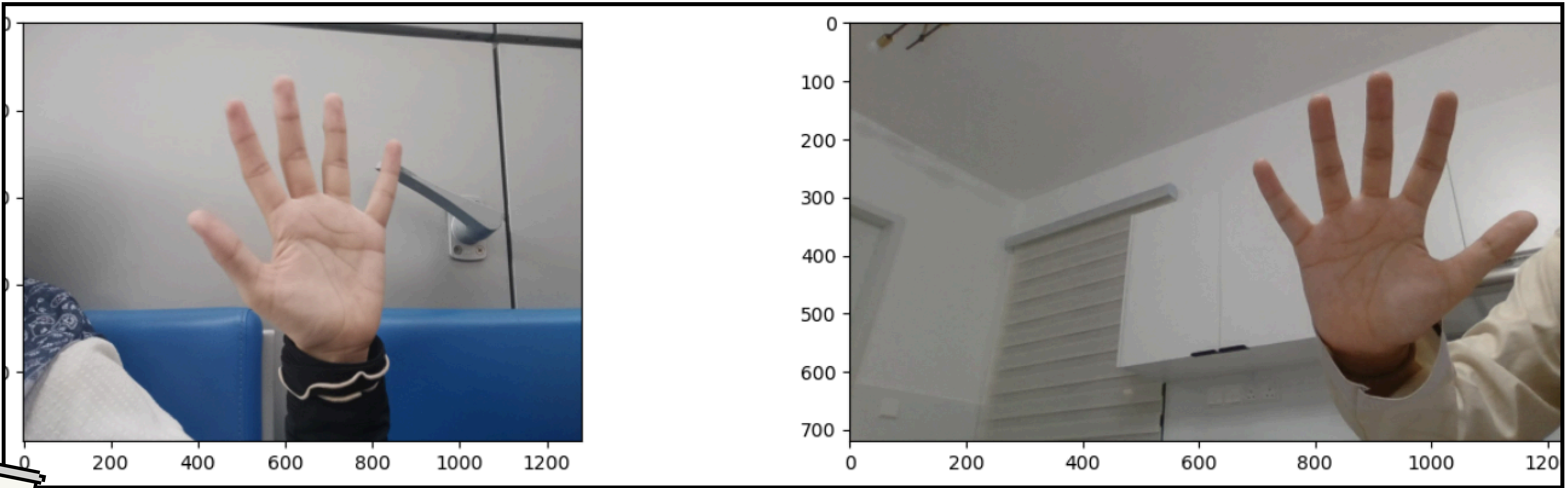
- ensures model can recognize gestures in different backgrounds



EDA KEY FINDINGS (3/3)

Position variation

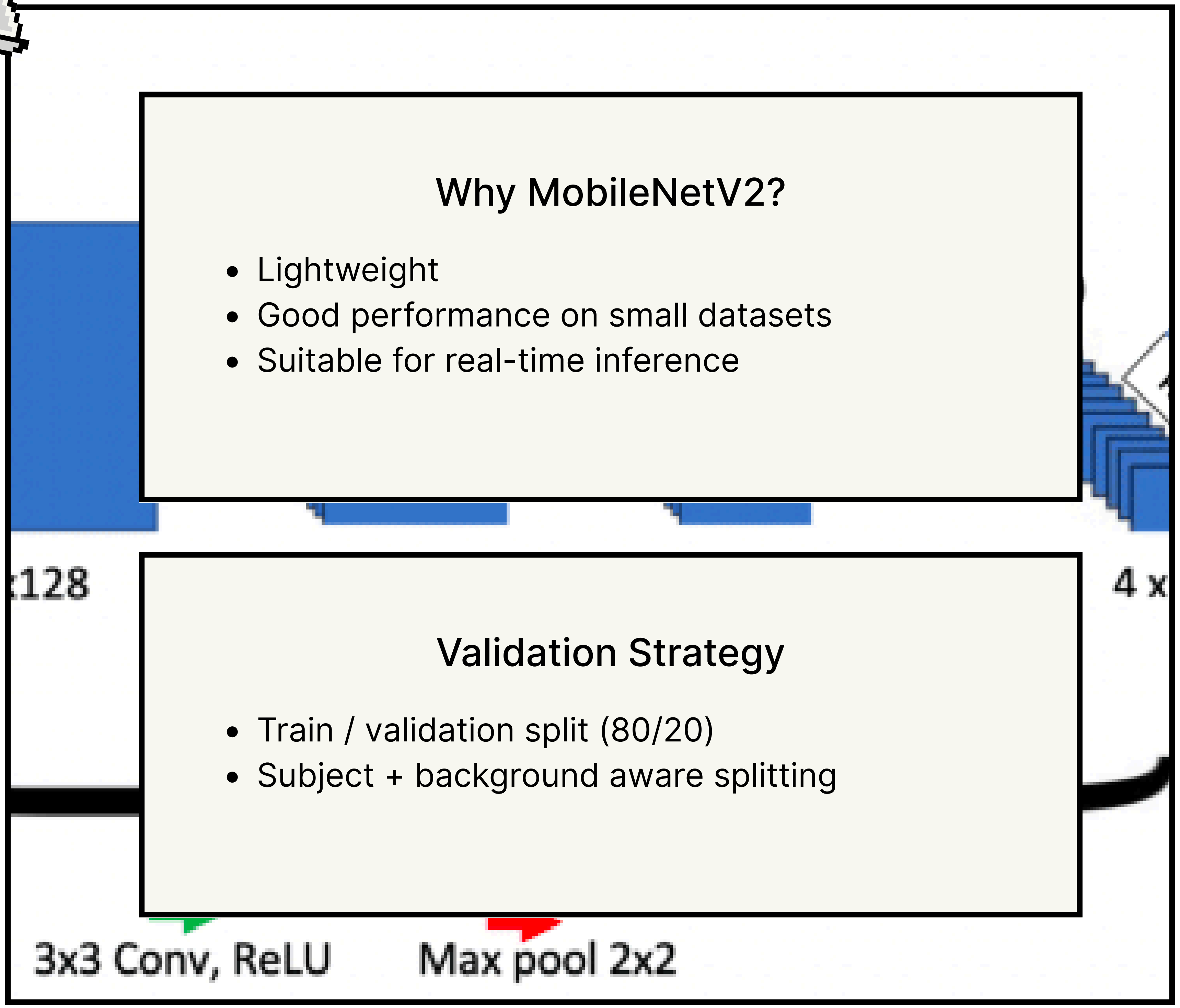
- ensures model can recognize gestures in different positions



MODELING APPROACH

Models tried:

- Custom CNN
- MobileNetV2 (transfer learning)



Why MobileNetV2?

- Lightweight
- Good performance on small datasets
- Suitable for real-time inference

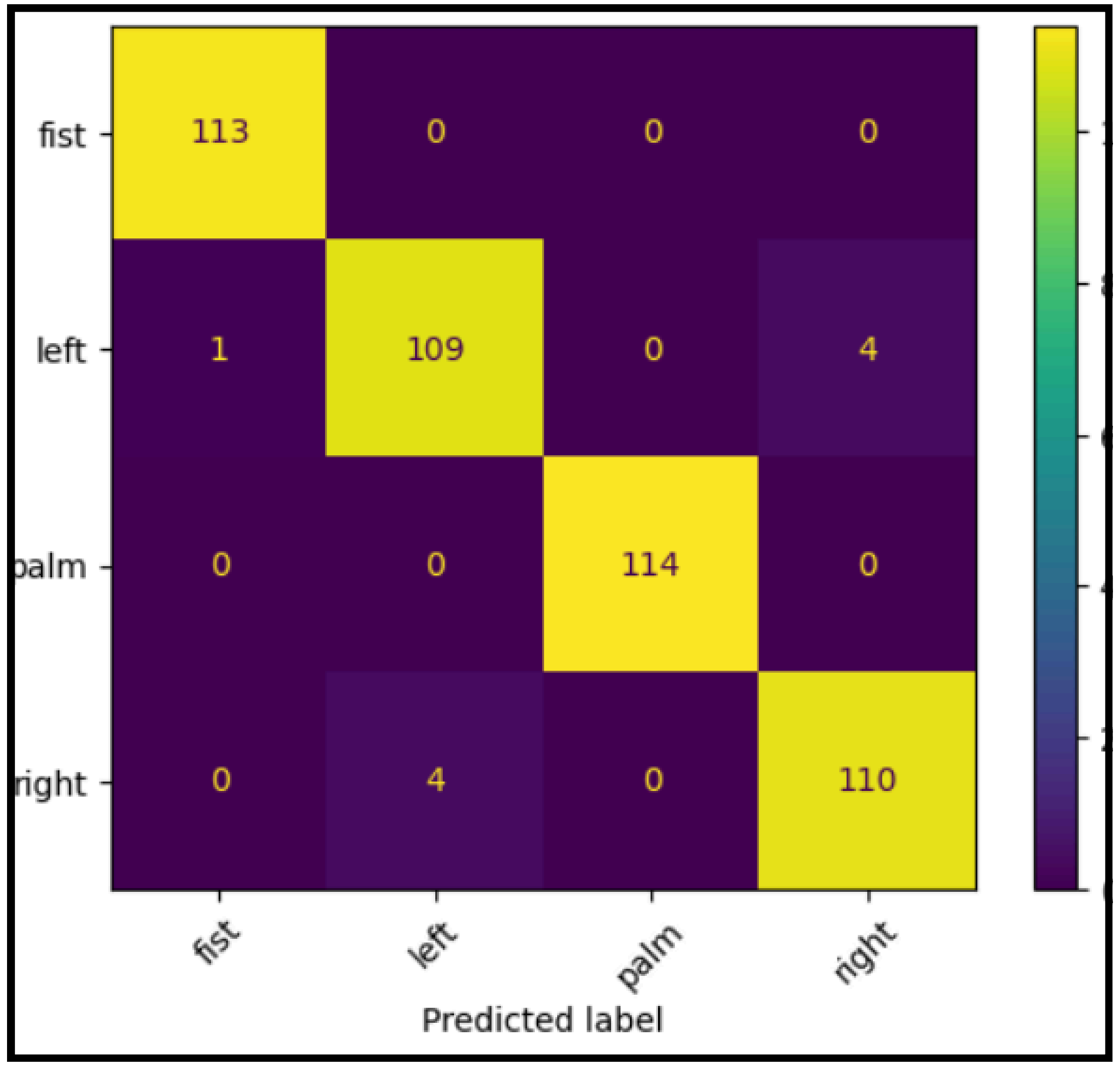
Validation Strategy

- Train / validation split (80/20)
- Subject + background aware splitting

RESULT & EVALUATION

Metrics:
- Validation accuracy
- Confusion Matrix
- Test on sample images

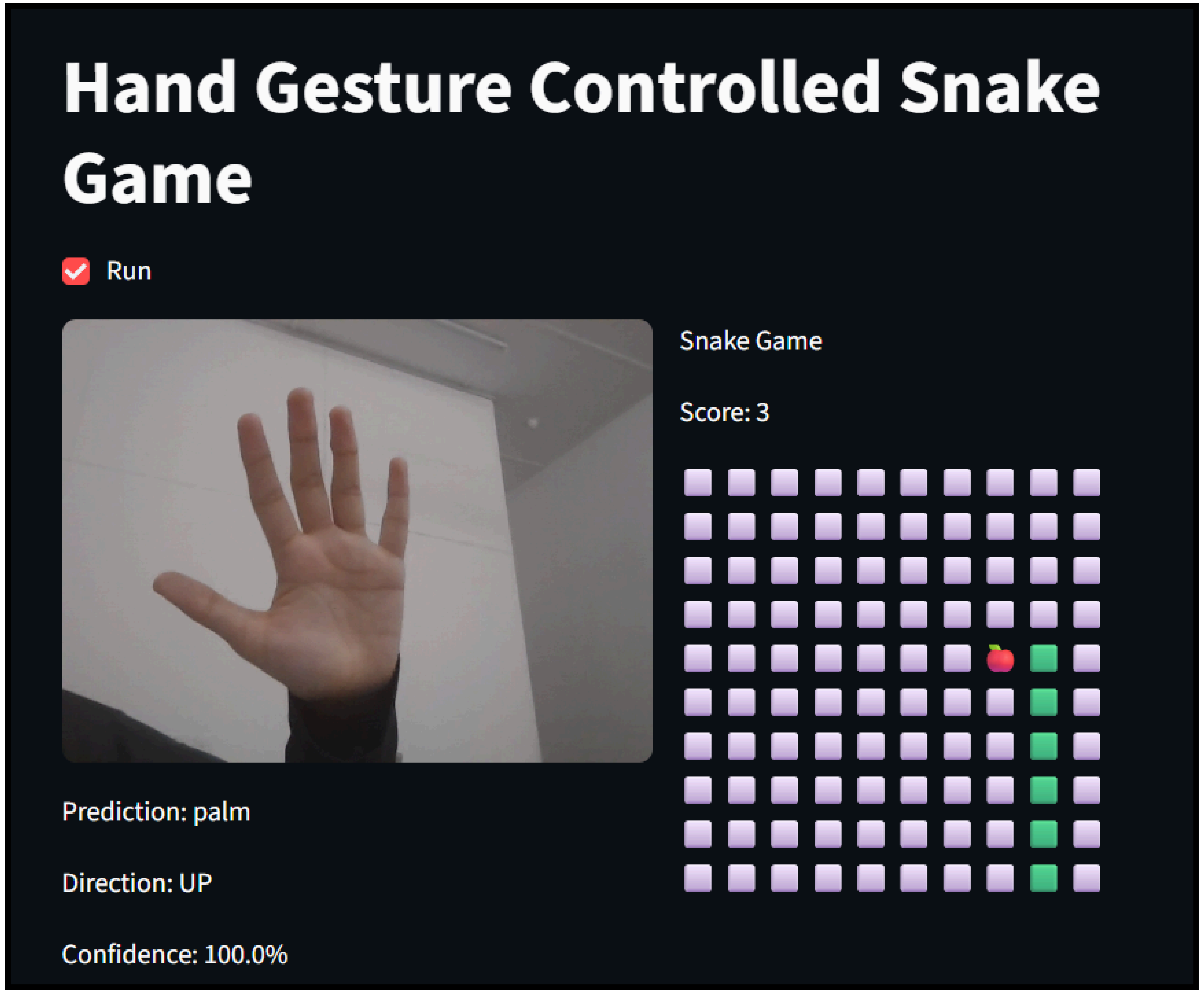
```
Found 455 images belonging to 4 classes.
15/15 ━━━━━━━━━━━━━━━━━━━━ 11s 613ms/step
Validation Accuracy: 0.9802
```



PROJECT DEMO

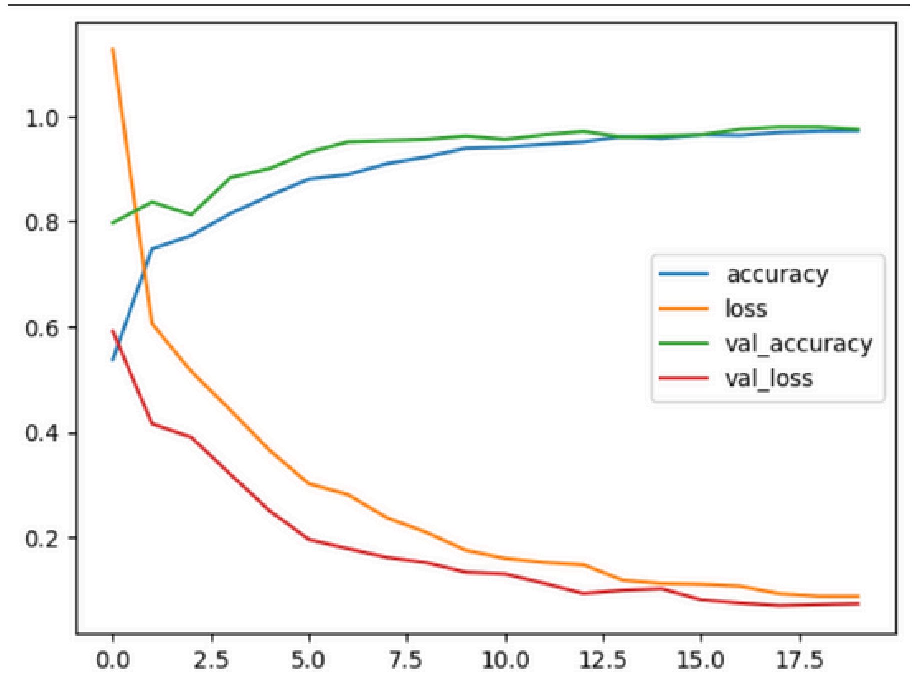
Input (frame)

Output (prediction)



Snake coordinates change according to output

MEASURES OF SUCCESS




```
Found 455 images belonging to 4 classes.
15/15 ━━━━━━━━━━━━━━━━━━━━ 11s 613ms/step
Validation Accuracy: 0.9802
```

Technical KPI

- Target: Accuracy > 90%
- Achieved: ~98%

Practical KPI

- Working real-time prediction from webcam frame
- Successful gesture-controlled gameplay

CHALLENGES / LIMITATIONS

Small Dataset
(~2500 imgs)

Left-right
classification
Confusion

Confusion in cases
where hand is off
center

FUTURE WORK & RECOMMENDATIONS

More diverse data (subjects, environment, etc.)

Data collection automation

Replace point left/right with more recognizable gestures

Use model in other games

TECH STACK

Language

- Python

Libraries

- TensorFlow
- Keras
- NumPy
- Matplotlib

Infrastructure

- Git LFS
- GitHub
- Streamlit

THANK YOU FOR LISTENING

ANY QUESTIONS?

+123-456-7890

www.ainul.com

ainul@ainul.com